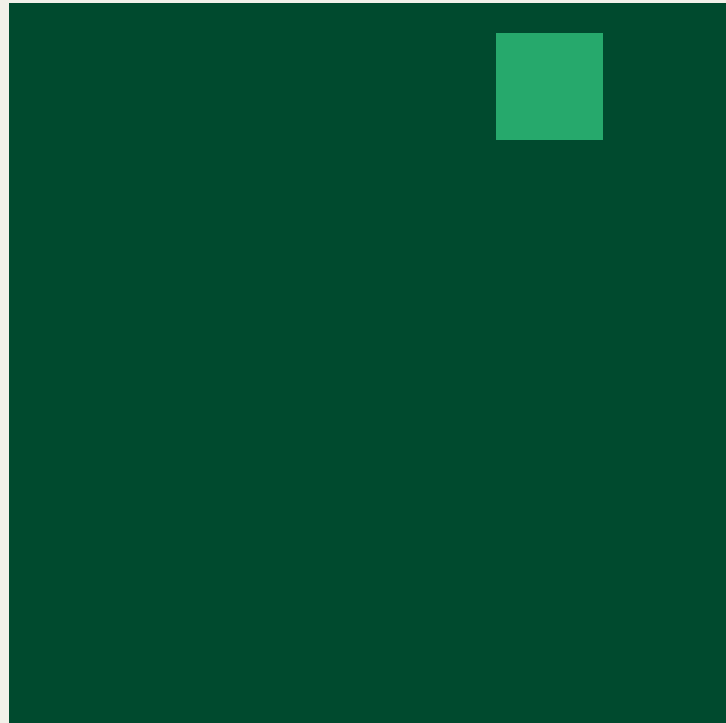


1.CNAME Resolution

2. RCODE Translation

Aaditya Deshmukh & Guojian Chen



Overview



■ BACKGROUND

- Built an iterative DNS resolver
- Phase 1: basic DNS query and response parsing
- Phase2: TTL-base DNS caching
- Phase3: CNAME resolution and improved error handling

■ Phase 3 Objectives

- Resolve domains that use CNAME aliases
- Report specific DNS response errors
- Validate response transaction IDs

Problem Statement: CNAME

Reference

RFC 1034 Section 3.6.2 - Aliases and canonical names

RFC 1035 Section 3.2.1 - Resource record TT

■ The Problem

Phase 2 only looked for direct A records

Some valid domains return a CNAME alias instead

Without CNAME support, the resolver could fail even though the domain exists

■ The Solution

Detect CNAME records in the DNS answer section.

Follow the alias until an A record is found.

Prevent infinite loops by limiting the chain length.

■ How we Address it

Added support for record type 5 (TYPE_CNAME).

Parsed CNAME data as another DNS name.

Recursively resolved each CNAME target.

Added loop detection and a maximum chain length of 10.

Cached CNAME records for reuse.

Problem Statement: Generic DNS Errors

Reference

RFC 1035 Section 4.1.1 - DNS header, transaction ID, and RCODE

■ The Problem

DNS responses use the **RCODE** field to identify errors.

Phase 2 treated different failures as the same generic error.

The resolver also did not verify that the response transaction ID matched the original query.

■ The Solution

Read the RCODE from the DNS response header.

Return specific errors such as:

- NXDOMAIN
- SERVFAIL
- FORMERR
- REFUSED

Reject responses with the wrong transaction ID.

■ How we Address it

Extracted the lowest four bits of the flags field.

Added an error-message mapping for each supported RCODE.

Compared packet.header.id with the original query ID.

Validated the response before caching or returning records.

Results

CNAME Resolution

```
kofan@GJ_desktop MINGW64 ~/OneDrive/Documents/GitHub/158A-GroupProject/Phase3 (aaditya-phase3)
$ python resolver.py www.microsoft.com
```

```
Resolving www.microsoft.com
[CACHE MISS] www.microsoft.com type=1
Querying 198.41.0.4 for www.microsoft.com
[NETWORK QUERY #1] 198.41.0.4 for www.microsoft.com
Querying 192.41.162.30 for www.microsoft.com
[NETWORK QUERY #2] 192.41.162.30 for www.microsoft.com
Querying 150.171.10.39 for www.microsoft.com
[NETWORK QUERY #3] 150.171.10.39 for www.microsoft.com
[CNAME] www.microsoft.com -> www.microsoft.com-c-3.edgekey.net
[CACHE MISS] www.microsoft.com-c-3.edgekey.net type=1
Querying 198.41.0.4 for www.microsoft.com-c-3.edgekey.net
[NETWORK QUERY #4] 198.41.0.4 for www.microsoft.com-c-3.edgekey.net
Querying 192.55.83.30 for www.microsoft.com-c-3.edgekey.net
[NETWORK QUERY #5] 192.55.83.30 for www.microsoft.com-c-3.edgekey.net
Querying 23.211.133.65 for www.microsoft.com-c-3.edgekey.net
[NETWORK QUERY #6] 23.211.133.65 for www.microsoft.com-c-3.edgekey.net
[CNAME] www.microsoft.com-c-3.edgekey.net -> e13678.dscb.akamaiedge.net
[CACHE MISS] e13678.dscb.akamaiedge.net type=1
[CACHE REFERRAL] zone=net, server=m.gtld-servers.net, ip=192.55.83.30
Querying 192.55.83.30 for e13678.dscb.akamaiedge.net
[NETWORK QUERY #7] 192.55.83.30 for e13678.dscb.akamaiedge.net
Querying 184.26.161.192 for e13678.dscb.akamaiedge.net
[NETWORK QUERY #8] 184.26.161.192 for e13678.dscb.akamaiedge.net
Querying 23.200.73.222 for e13678.dscb.akamaiedge.net
[NETWORK QUERY #9] 23.200.73.222 for e13678.dscb.akamaiedge.net
23.37.18.39
```

Error Handling

```
kofan@GJ_desktop MINGW64 ~/OneDrive/Documents/GitHub/158A-GroupProject/Phase3 (aaditya-phase3)
$ python resolver.py nonexistent-example-test-5.com
```

```
Resolving nonexistent-example-test-5.com
[CACHE MISS] nonexistent-example-test-5.com type=1
Querying 198.41.0.4 for nonexistent-example-test-5.com
[NETWORK QUERY #1] 198.41.0.4 for nonexistent-example-test-5.com
Querying 192.41.162.30 for nonexistent-example-test-5.com
[NETWORK QUERY #2] 192.41.162.30 for nonexistent-example-test-5.com
DNS error: NXDOMAIN: the domain name does not exist
```

Thank You!

